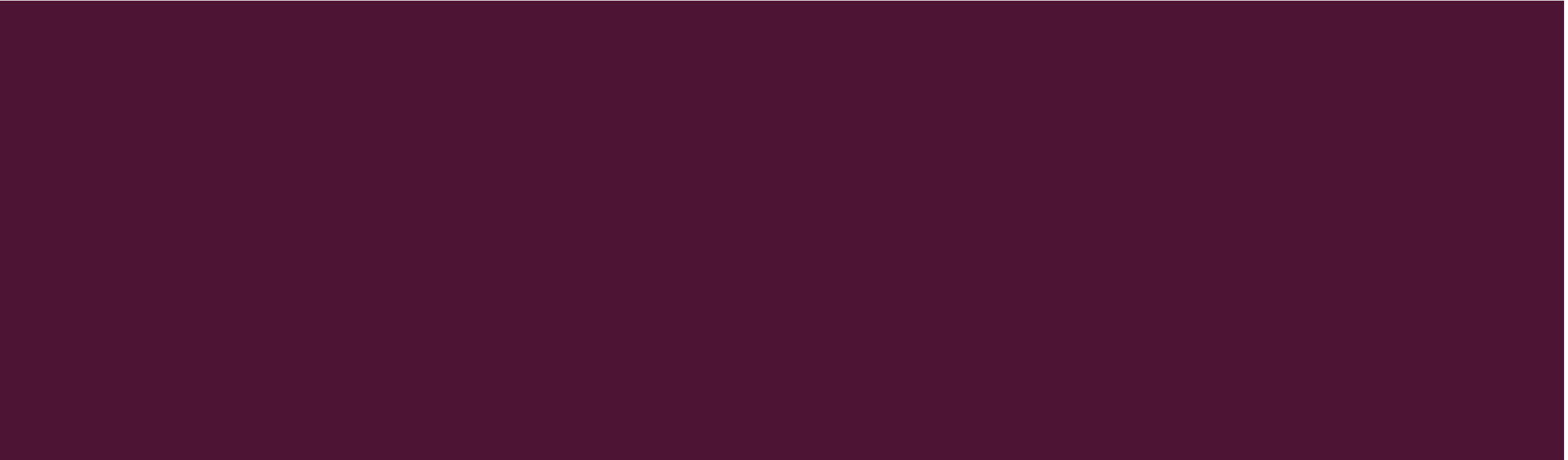




LECTURE ON IOT (PART 2)

RFID & NFC



What is RFID

Application of RFID

What is inside in RFID

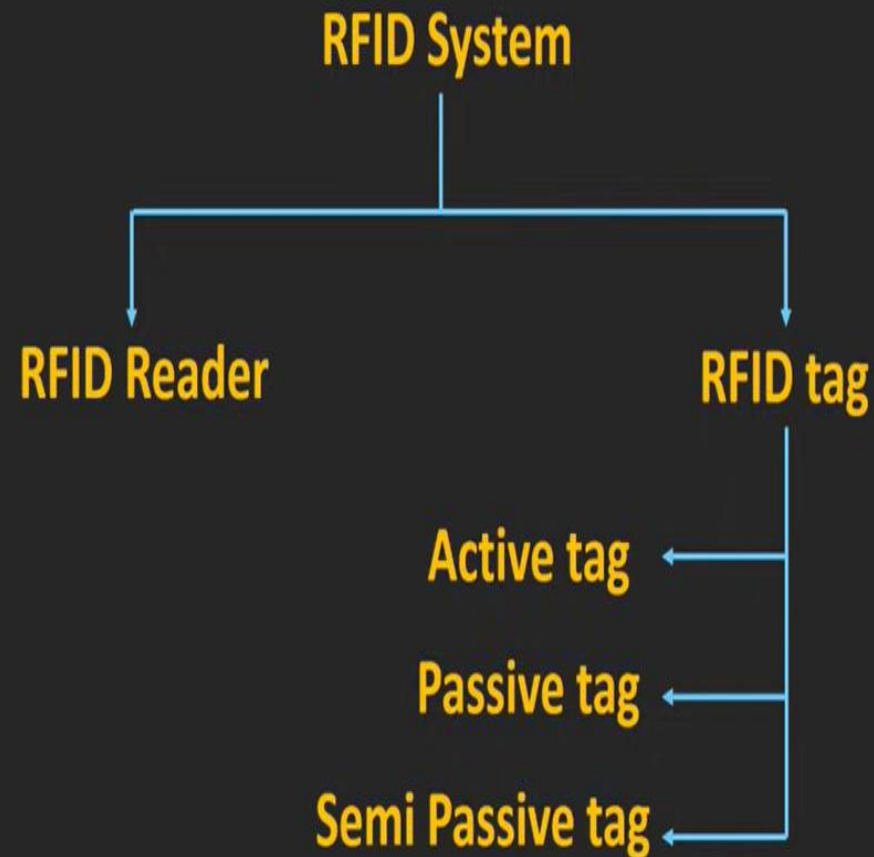
How RFID Works? (Operating Principle)

What is RFID



RFID stands for **Radio Frequency Identification**. It is a wireless technology used to **identify, track, and manage objects, animals, or people** using radio waves.





1. Passive RFID Tag

- A tag that has **no internal battery** and is powered by the radio waves from the RFID reader. Cheaper, smaller, and has shorter read range (a few cm to a few meters). Common in ID cards, library books, item tracking.

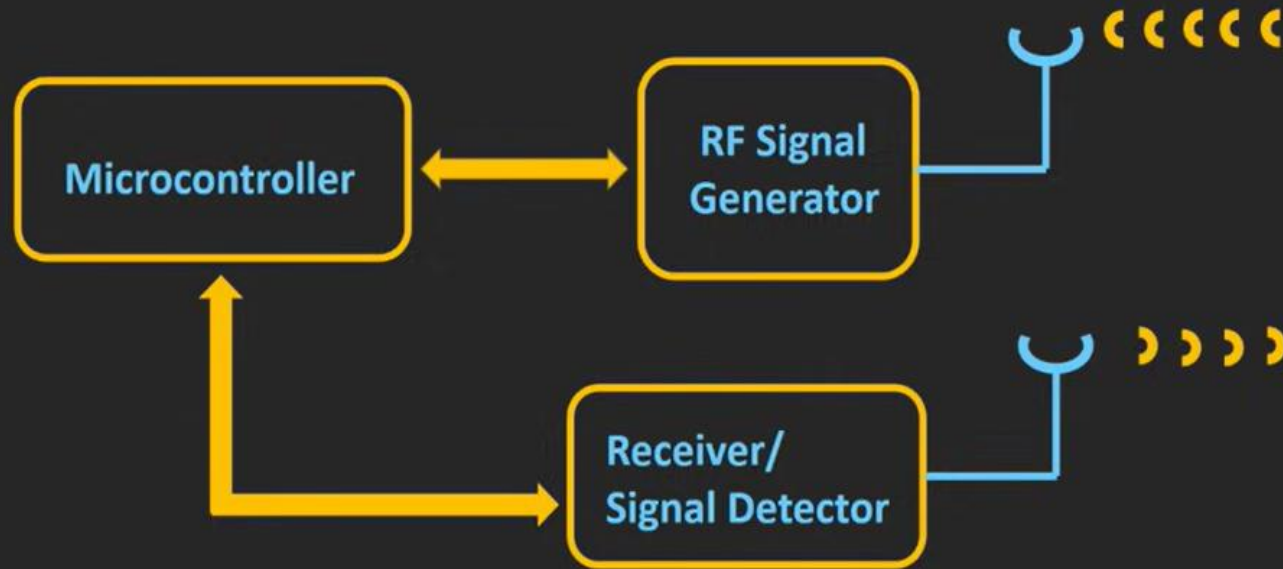
2. Active RFID Tag

- A tag that has its **own battery** and actively sends signals to the reader. Longer read range (tens to hundreds of meters), more expensive, used for vehicle tracking, toll booths, and large asset monitoring.

3. Semi-Passive (Battery-Assisted Passive) RFID Tag

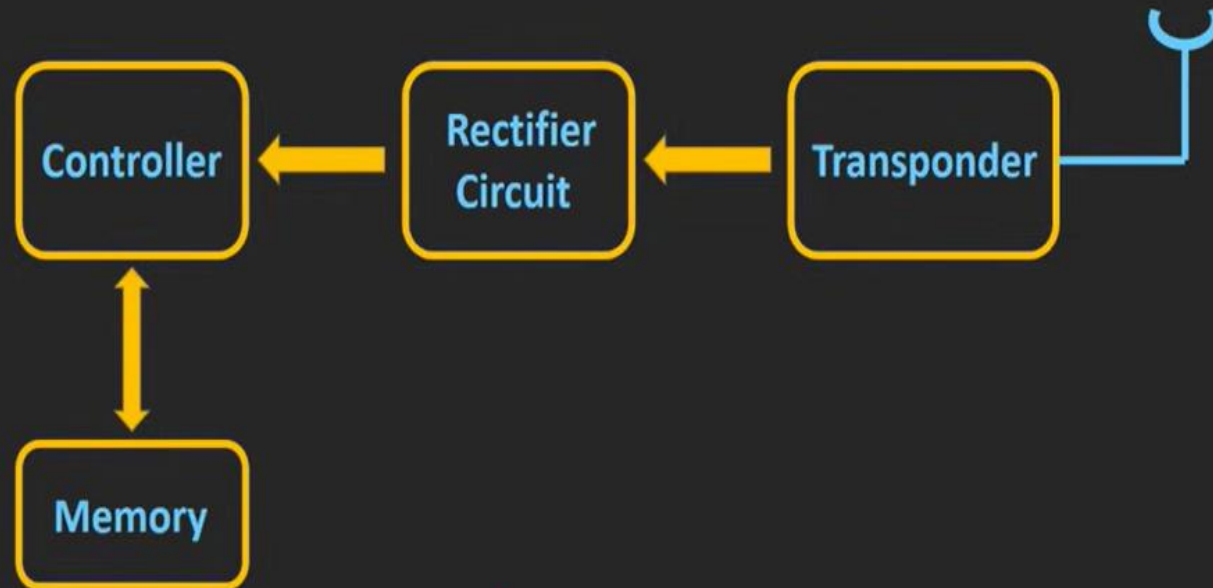
- A tag that has a **battery** but only uses it to power the internal circuit, not for transmitting signals. The reader still triggers communication, but battery improves sensitivity and range.

RFID Reader



- The **signal generator** produces radio waves.
- The **antenna** sends these waves to the RFID tag.
- The **detector** receives the signal reflected or modulated by the tag.
- The **microcontroller** processes the detected signal and extracts the tag's information.

RFID Tag



- **Rectifier Circuit:** Converts the **RF signal from the reader** into DC power to activate the tag.
- **Controller:** Manages the tag's operations, including receiving commands and controlling data flow.
- **Memory:** Stores the tag's unique ID or other information that needs to be sent to the reader.
- **Transponder:** Modulates and also sends the stored data back to the reader

OPERATING FREQUENCY RANGE

Frequency of Operation

```
graph TD; A[Frequency of Operation] --> B[LF]; A --> C[HF]; A --> D[UHF]; B --> B1["(Low Frequency)"]; B1 --> B2["125 kHz or 134 kHz"]; B2 --> B3["Range: up to 10 cm"]; C --> C1["(High Frequency)"]; C1 --> C2["13.56 MHz"]; C2 --> C3["Range: up to 1 m"]; D --> D1["(Ultra High Frequency)"]; D1 --> D2["860- 960 MHz"]; D2 --> D3["Range: 10 to 15 m"];
```

LF

(Low Frequency)
125 kHz or 134 kHz

Range: up to 10 cm

HF

(High Frequency)
13.56 MHz

Range: up to 1 m

UHF

(Ultra High Frequency)
860- 960 MHz

Range: 10 to 15 m

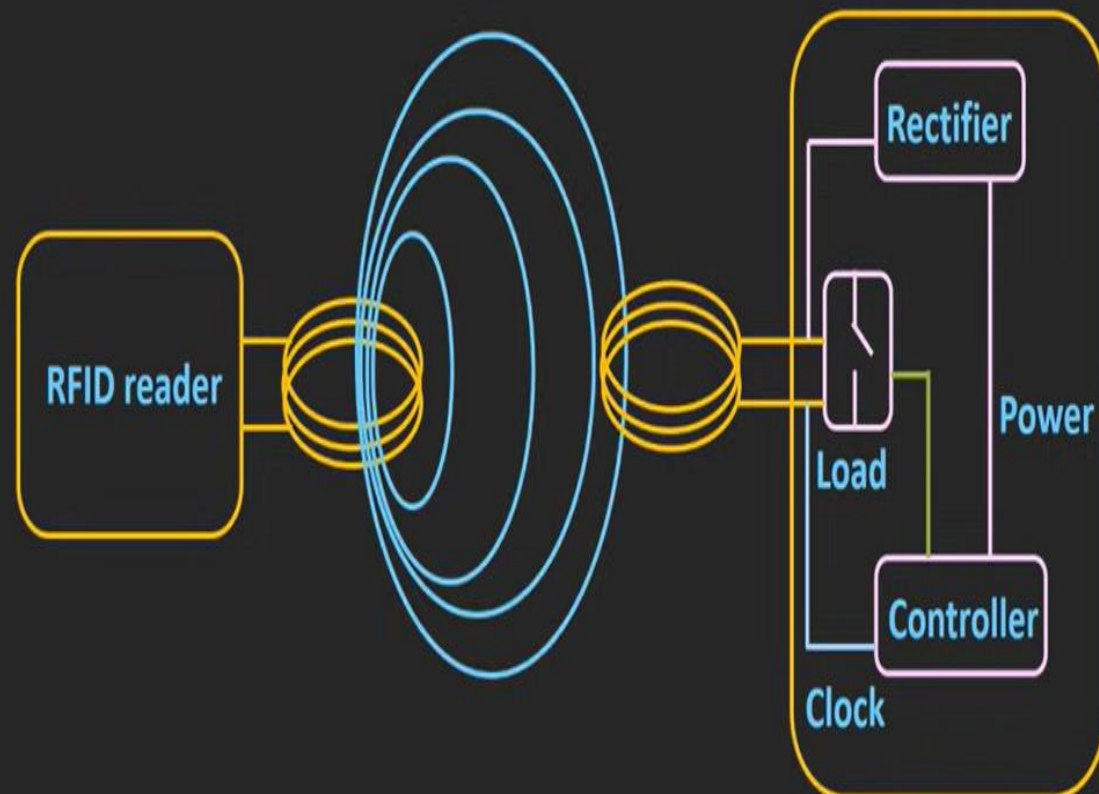
WORKING PRINCIPLE

LF and HF RFID Tags: **Inductive Coupling (Near Field Coupling)**

UHF RFID Tags: **Electromagnetic Coupling (Far Field Coupling)**

NEAR FIELD COUPLING

Inductive Coupling (Near Field Coupling)

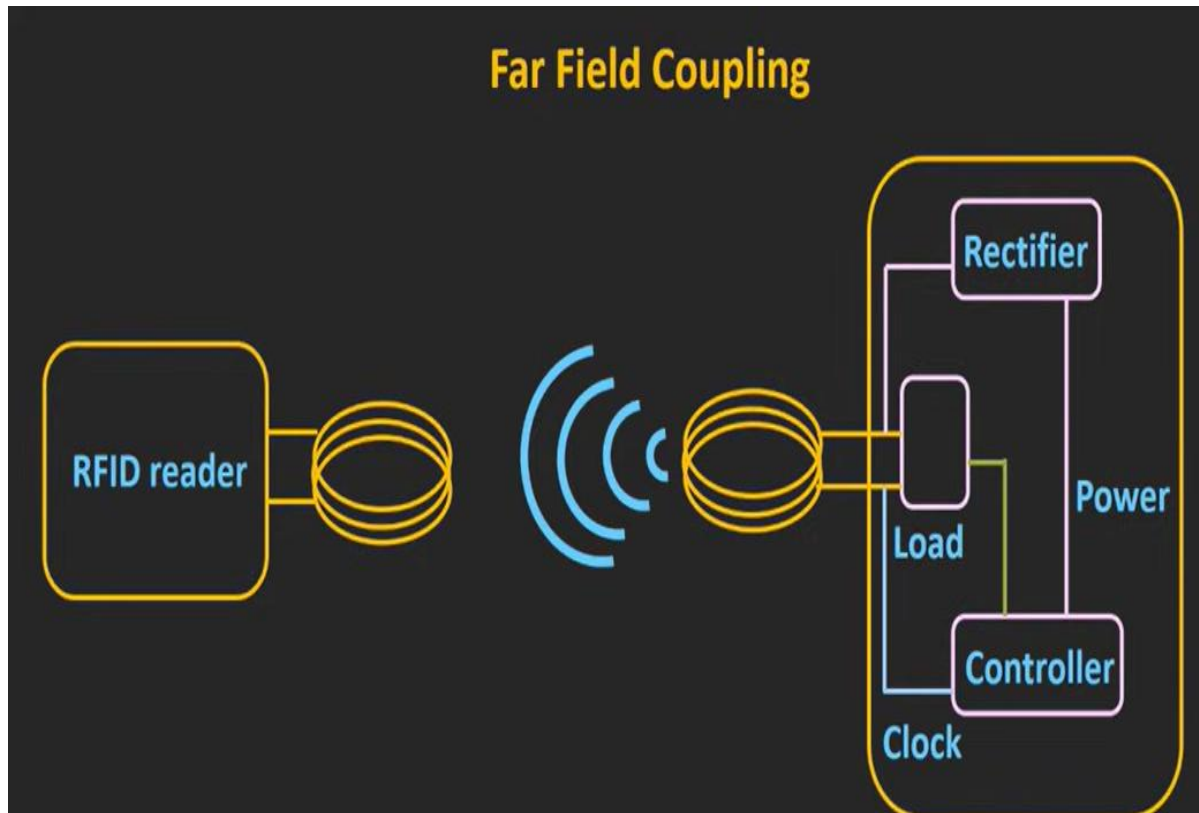


The RFID **reader** has a coil antenna that produces an **alternating magnetic field** when current flows through it. This magnetic field extends only a **short distance** (a few centimeters to a meter). The **RFID tag** also has a coil (like a small loop). When the tag enters the reader's magnetic field, a **voltage is induced** in the tag's coil. This is called **electromagnetic induction**. The induced voltage is **converted to DC** using the tag's **rectifier circuit**. This **powers the tag's controller and memory**, so it can operate without its own battery (for passive tags). The tag **modulates the load** on its coil (changes its impedance) according to the data in its **memory**. This modulation slightly affects the magnetic field of the reader. The **reader detects these changes** and decodes the tag's information.

LOAD MODULATION

- The tag “tells” the reader its information by **slightly altering the magnetic field** from the reader.
- When a passive RFID tag **changes the load on its coil** to send data back to the reader.

FAR FIELD COUPLING



The **reader emits UHF radio waves** that travel through space. The waves can travel **long distances** (several meters). The **tag receives energy** from the waves to power itself (for passive tags). The **tag's antenna captures the waves** and gets ready to respond. The **tag reflects the waves back** by changing its antenna's load (**backscatter modulation**). This reflection **carries the tag's stored data**. The **reader's antenna receives the reflected waves**. The **reader detects the changes** and **decodes the information** from the tag.

NFC - FFC

Feature	NFC (Near Field)	FFC (Far Field)
Coupling Type	Magnetic (inductive) coupling	Electromagnetic (radio wave) coupling
Frequency	LF/HF (125 kHz – 13.56 MHz)	UHF (860–960 MHz)
Range	Very short (a few cm)	Longer (meters)
Energy Transfer	Reader's magnetic field powers passive tags	Tag uses backscatter to reflect reader's waves
Data Transfer Method	Load modulation (tag changes coil load)	Backscatter modulation (tag changes antenna reflection)
Speed	Slower	Faster
Applications	Contactless payments, access cards, NFC phones	Supply chain tracking, toll collection, UHF RFID systems



THANK YOU

P R E S E N T A T I O N T E M P L A T E

